

August 28, 2026

Targeted Talent

Dear Hiring Team,

I am writing to apply for the Data Scientist position with Targeted Talent's client. I recently completed my M.Sc. in Computer Science at the University of Calgary, and this role is especially interesting to me because it combines Python, machine learning, SQL/NoSQL data extraction, and practical modelling work around supply and demand problems.

My graduate research focused on machine-learning and data-management systems for large geospatial and Earth observation datasets. I developed Python and C++ pipelines for data collection, preprocessing, validation, storage, retrieval, model training, and performance evaluation. This work required careful attention to data quality, sampling strategy, scalability, and computational efficiency, because small preparation decisions could strongly affect model behaviour.

One of my main projects examined the reliability and generalizability of deep-learning models for digital elevation model super-resolution. I built workflows using Python, NumPy, pandas, Rasterio, and PyTorch; trained convolutional neural networks; and evaluated performance with statistical and terrain-aware metrics. I identified how random data splitting could make results look better than they would be in practice, then used region-based validation for a more realistic assessment.

My second major research area involved a DGGS-based framework for managing and retrieving large volumes of spatiotemporal satellite data. I designed multiresolution storage and retrieval methods, implemented performance-sensitive C++ components, and evaluated the system using large Canada-wide datasets. This gave me practical experience with scale, data architecture, reproducible experimentation, and making complex datasets usable for downstream analysis.

I also bring backend and database experience from Asr Gooyesh Pardaz, where I developed Django REST APIs backed by PostgreSQL for NLP, text-to-speech, and speech-to-text products. I implemented service-plan logic, authentication features, payment-service integration, and relational records for plan and payment workflows. This helps me connect modelling and analysis work with production-style data systems, APIs, and business processes.

The role's focus on high-quality data science models and supply/demand balancing is a strong fit for the way I like to work: understand the data carefully, build a defensible modelling pipeline, test assumptions, and communicate tradeoffs clearly. My strongest tools are Python, PyTorch, pandas, NumPy, SQL, PostgreSQL, Git, Linux, and scientific computing. As a teaching assistant in big data, data-at-scale, and database courses, I also learned to explain technical work clearly to different audiences.

I am currently based in Calgary and meet the Canadian location requirement. The remote-to-start structure is attractive to me, and a later Calgary path would be a natural fit. I would be excited to bring my machine-learning, large-scale data, Python, and database experience to this role while growing in applied data science work with direct business impact.

Thank you for considering my application. I would welcome the opportunity to discuss how my background in data science, ML model evaluation, scalable data systems, and technical communication can contribute to your client's team.

Sincerely,
Amir Mirzai Golpayegani
amirgolpaa24@gmail.com